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MILITARY SPECIFICATION

INDICATOR, TRUE AIRSPEED, TYPE M-1A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of pitot-static true airspeed indicator, range 100-450 knots, designated type M-1A.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-636	Box Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging
MIL-G-81704	Glass, Aircraft Instrument, Lighting Wedge and Cover

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STANDARDS

Federal

Federal Standard No. 595 Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking of Shipment and Storage
MIL-STD-130	Identification Marking of U S Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS28036	Indicator - True Airspeed, Type M-1A
MS28105	Cover Glass, Aircraft Instrument Dial
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Qualification. The indicators furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids. (reference paragraph 4.2 and 6.4)

3.2 Data. Unless otherwise specified in the contract or order, no data (other than reports and drawings accompanying qualification samples) are required by this specification or any of the documents referenced in section 2 (see 6.2).

3.3 Materials.

3.3.1 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

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3.3.2 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.3.2.1 Standard parts. Standard parts (MS or AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts, such as screws, bolts, nuts, and cotter pins may be used, provided they possess suitable properties and are replaceable by the standard parts (MS or AN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.4 Design and construction. The indicator shall be as shown on MS28036. The indicator shall be so constructed that no part will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Temperature and pressure compensation. The indicator shall be so designed that proper compensation is made for variations in pressure and temperature.

3.4.2 Case. The indicator case shall be as shown on MS28036. The case shall be made either of nonferrous low density metal or synthetic material, uniform in texture with a smooth surface, and entirely covered with a durable black finish. Synthetic material shall be of the thermal setting type, composed of a suitable filler and phenol condensation product binder.

3.4.2.1 Reinforcement. The inside of the case around each connection shall be of sufficient strength to prevent damage to the case when the connections are tightened during installation of the indicator. If a synthetic material is used in the construction of the case, one brass or aluminum alloy insert for the static pressure connection, one for the pitot pressure connection, and one for the free air temperature capillary tubing connection shall be molded in the case or suitably fastened by screws. If of aluminum alloy, the insert shall be anodically treated, except for the threads.

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3.4.2.2 Bezel ring. The bezel ring shall be made of nonferrous low density metal and shall have a durable dull black finish. The ring shall be held in place by means of screws properly secured by lock washers or similar devices of approved design. Provision shall be made for replacement of the cover glass by removal of the bezel ring.

3.4.2.3 Cover glass. The cover glass shall be MS28105-16, type I, with antireflection coating and shall be properly sealed. The distance between the cover and the dial shall be the minimum practicable and shall not exceed 0.187 inch.

3.4.2.4 Temperature capillary. The free air temperature capillary shall conform to the dimensions shown on MS28036. The sunshield and mounting assembly shall be removable so that the bulb may be inserted through a 3-1/4 inch hole in the airplane skin. The sunshield and mounting assembly shall have a smooth dull finish. Screws, nuts, washers, etc., for installing the sunshield and temperature bulb shall be furnished.

3.4.2.4.1 Location. The location of the connection of the free air temperature capillary tube to the indicator case shall conform to MS28036.

3.4.2.4.2 Length. The length of the free air temperature capillary shall be 8 feet, unless otherwise specified by the procuring activity. The capillary length shall be marked on the case.

3.4.2.5 Capillary tubing. The free air temperature capillary tubing shall be covered throughout its length with a flexible braided casing. The casing shall be securely attached to the case and to the bulb to protect the capillary tubing against strains.

3.4.2.6 Capillary reinforcement. A suitable flexible reinforcement, extending 4 inches from the case and from the bulb, shall be provided on each end of the free air temperature capillary casing. The reinforcement shall be of a flexibility to withstand, without failure, a 180-degree bend of 1-3/8 inch radius. The reinforcement shall be securely attached to the case and to the bulb. The inside diameter of the reinforcement shall be such that the reinforcement will slide freely on the capillary casing. The reinforcement shall be braided brass or bronze spiral armor, approximately 9/32 inch outside diameter, plain weave, seven strands. The brass or bronze wire shall be 0.011 inch in diameter with 6-1/2 to 7 strands per inch.

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3.4.2.7 Cleaning. The free air temperature capillary tubing. shall be thoroughly cleaned at the proper stage of assembly and shall be filled in a manner to exclude air and other non-condensing gases and moisture.

3.4.2.8 The back of the case, adjacent to the pressure connections, shall be marked as follows:

"P" Pressure connection
"S" Static connection

3.4.2.8.1 The following shall be marked on the back of the case:

CAUTION
DO NOT BLOW IN TUBES

3.4.3 Pointer. The MS33585-3 pointer shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration. The shaded portion of the pointer shall be finished in lusterless white, color No. 37875 of Federal Standard No. 595. The pointer shall be firmly attached to the mechanism, and yet be readily adjustable.

3.4.3.1 Limitation of pointer movement. The pointer movement shall be limited by suitable stops in the mechanism in such a way that the pointer will not be permitted to rotate more than 350 degrees of arc clockwise from the zero graduation when the indicator is submitted to overpressure tests. Suitable stops may also be incorporated in the indicator mechanism to limit the counterclockwise rotation of the pointer to the zero graduation. The design of the stops shall be subject to the approval of the procuring activity.

3.4.4 Instrument dial. The instrument dial shall conform to figure 1. The style and proportions of numerals and letters placed on the dial shall conform to MS33558. Numerals shall distinctly indicate the graduation to which each applies.

3.4.4.1 Markings. The following markings shall be finished in lusterless white, color No. 37875 of Federal Standard No. 595. The dimensions of the markings shall be as follows:

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	Height or length (inch) ±0.016	Width of line or graduation (inch) ±0.005
Numerals 100 to 400, as shown on figure 1	0.312	0.031
100-knot graduations	(figure 1)	.031
50-knot graduations	(figure 1)	.031
5- and 10-knot graduations from 100 to 500 knots, inclusive, as indicated on figure 1	(figure 1)	0.016
Lettering "TRUE"	0.188	.031
Zero numeral	.156	.016
Zero graduation	.125	.016
Lettering "KNOTS" and "AIRSPEED"	.125	.016

3.4.4.2 Durable dull black. The marking "TYPE M-1A" shall be durably and legibly marked on the dial in approximately 0.047-inch letters. These markings and all other markings not otherwise specified shall be finished in durable dull black color No. 37038 of Federal Standard No. 595.

3.4.4.3 Visibility of dial. The pointer, numerals, at least 0.062 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the instrument case.

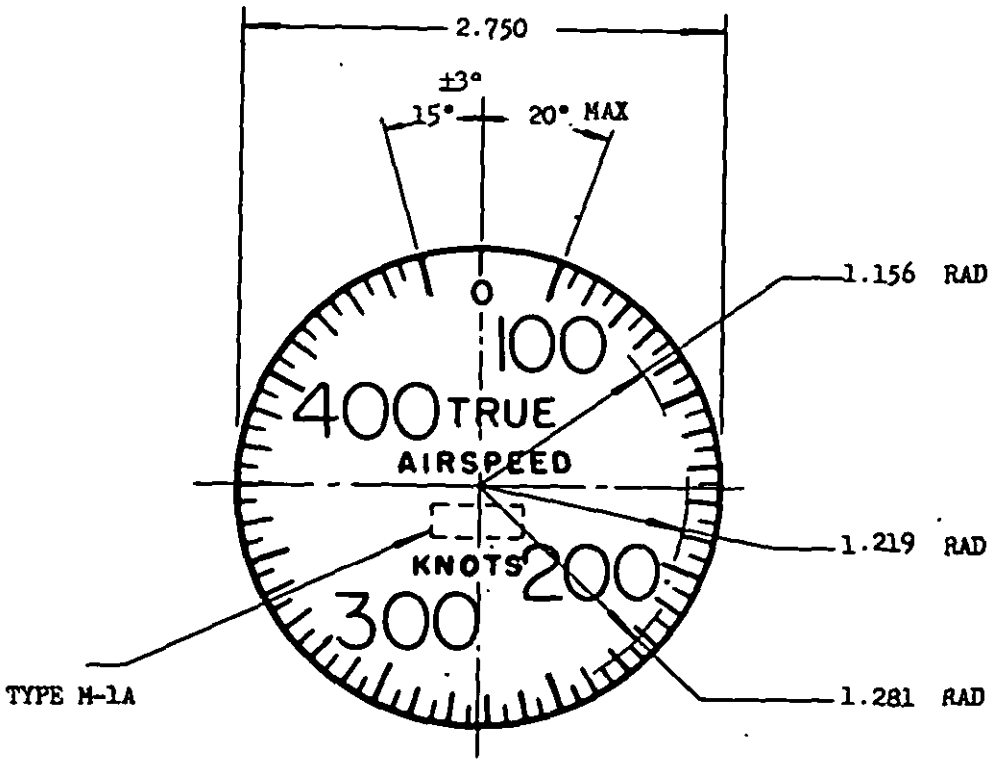
3.4.4.3.1 The background of the dial, unshaded part of the pointer, and the front of the bezel ring shall be finished in durable dull black color No. 37038 of Federal Standards No. 595.

*3.5 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.6 Application of pressure. The pitot pressure shall be applied to the inside of the diaphragm capsule, and the static pressure to the interior of the case.

3.7 Threads.

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UNLESS OTHERWISE SPECIFIED, DIMENSIONS IN INCHES. TOLERANCES: DECIMALS ±.0156

FIGURE 1. Instrument dial

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3.7.1 Screw threads. Unless otherwise specified, the threads of all machine screws, 0.060 inch or larger in diameter, shall conform to MIL-S-7742.

3.7.2 Pipe threads. Pipe threads shall conform to MIL-P-7105.

3.8 Weight. The weight of the indicator completely assembled, including mounting nuts, mounting screws, and sunshield, shall not exceed 3 pounds.

3.9 Finishes and protective coatings.

3.9.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows.

3.9.1.1 Dials, small holes, and case inserts need not be anodized.

3.9.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.9.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.9.1.4 Castings containing non-aluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541, in lieu of anodizing.

3.9.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.9.2 Steel parts. Steel parts shall be cadmium plated, where practicable, in accordance with QQ-P-416, types II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.10 Performance. The indicator shall be capable of meeting the performance requirements specified in section 4.

3.11 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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3.11.1 Use of MS or AN designations. MS or AN designations shall not be applied to a product, except for qualifications test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for qualification.

3.12 Installation.

3.12.1 Mounting screws. The contractor shall furnish sufficient mounting screws for installing the indicator. They shall be No. 6-32 NC-2 round-head brass machine screws having a durable dull black finish. The length shall be sufficient to permit mounting on a panel 3/16 inch thick.

3.12.2 Accessories. The contractor shall furnish with each indicator a sunshield, reinforcing ring, split clamping ring, gasket, and sufficient mounting screws and nuts, conforming to the dimensions shown on MS28036.

3.12.3 Envelope. An envelope, containing mounting screws and accessories shall be furnished with each indicator. The following information shall be printed on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS AND ACCESSORIES

3.13 Workmanship.

3.13.1 General. The indicator, including all parts and accessories, shall be finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of welding, brazing, painting, alinement of parts and tightness of assembly screws and bolts; etc.

3.13.2 Dimensions. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.13.3 Screw assemblies. Assembly screws and bolts shall be tight. The word "tight" means that the screw or bolt cannot be appreciably tightened further, without damage to the screw or bolt or threads.

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3.13.4 Cleaning. The indicator shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign material, after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of tests. The inspection and testing of the indicator shall be classified as:

(a) Qualification inspection (4.4)
(b) Quality conformance inspection (4.5)

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of test are not specified definitely it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Tapping. Unless otherwise specified, the indicator shall be gently tapped or vibrated before a test reading is taken.

4.3.3 Attitude. Unless otherwise specified, the indicator shall be tested in the normal operating position.

4.4 Qualification inspection.

4.4.1 Prior qualification. Unless otherwise specified by the procuring activity, indicators which have not previously passed a qualification test or which have passed the qualification tests and have been modified in any manner, shall satisfactorily pass a qualification test prior to the acceptance of any indicator.

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4.4.2 Sampling instructions. Qualification test samples shall consist of three indicators. Samples shall be identified as required and forwarded to the activity responsible for qualification designated in the letter of authorization from that activity (see 6.4).

4.4.3 Tests. Qualification tests shall consist of all the tests specified herein.

*4.5 Quality conformance inspection. Quality conformance tests shall consist of individual tests, sampling tests, and special tests.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests. In addition, each indicator shall be subjected to any other test specified herein considered necessary to determine compliance with this specification.

4.5.1.1 Examination of product. Each indicator shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, and marking.

4.5.1.2 Scale error. The indicator shall be mounted on a panel that has facilities for tapping to remove the friction effect. The free air temperature bulb shall be placed in a free air thermometer test fixture in which it is possible to control, to vary, and to read the temperature accurately throughout the range of from -60° to +40°C.

4.5.1.2.1 The static pressure connection of the indicator shall be connected by means of a cross, to an accurate barometer, a source of air pressure and suction, and to the suction side of a water manometer.

4.5.1.2.2 The pitot pressure connection of the indicator shall be connected, by means of a tee, to a source of air pressure and to the pressure side of the water manometer.

4.5.1.3 Each indicator shall be tested for scale errors as follows:

4.5.1.3.1 The temperature around the free air temperature bulb in the test fixture shall be subjected to the first test setting specified in table I.

4.5.1.3.2 The static pressure shall be adjusted to the standard value (29.92 inches Hg) corresponding to zero feet altitude, sea level standard, as specified in table I.

4.5.1.3.3 The pitot pressure shall then be so adjusted that the indicated airspeed indicated by the manometer is equal to the first test point, as specified in table I.

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4.5.1.3.4 While the temperature of the free air temperature bulb is maintained at the first test setting, as specified in table I, and the static pressure adjusted to zero feet altitude, the pitot pressure shall be so varied that the differential pressure is equal to each of the values marked with an asterisk in column 2 of table I. Under these conditions, the true airspeed indicator shall indicate the true airspeeds listed in column 3 and shall not vary from these indications by more than the tolerances specified in 4.5.1.3.8. This procedure shall be followed for each of the differential pressure altitudes marked by an asterisk in table I.

4.5.1.3.5 The above procedure shall be repeated for the conditions specified in tables II, III, IV, V, and VI.

4.5.1.3.6 With pressures increasing, the pressure shall be brought up to but shall not exceed the pressures specified to give the desired reading, and with pressures decreasing, the pressure shall be brought down to but shall not fall below the pressure specified to give the desired reading. The errors at the test points shall not exceed the tolerances specified in 4.5.1.3.8.

TABLE I. Test settings at 40° C

Pressure altitude (feet)	Indicated airspeed (knots)	True airspeed (knots)
0	100	104
	150*	156
	200	207
	250*	258
	300	308
	350*	357
	400	405
	(450)	(453)
5,000	100	114
	150*	170
	200	226
	250*	281
	300	335
	350*	387
	400	438

*See paragraphs headed "Friction" and "Position error" (see 4.5.1.4 and 4.5.2.10). Tests at points marked with asterisks shall be made with the pressure increasing and then with pressure decreasing.
Values inclosed in parentheses are for information only.

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TABLE II. Test settings at 20° C

Pressure altitude (feet)	Indicated airspeed (knots)	True airspeed (knots)
0	100	101
	150*	151
	200	200
	250*	249
	300	298
	350*	345
	400	392
	450*	438
5,000	100	110
	150*	165
	200	219
	250*	272
	300	324
	350*	375
	400	424
	(450)	(472)
10,000	100	121
	150*	181
	200	239
	250*	297
	300	353
	350*	407
	(400)	(459)

*See paragraphs titled "Friction" and "Position error" (see 4.5.1.4 and 4.5.2.10). Tests at the points marked with asterisks shall be made with pressure increasing and then with pressure decreasing.

Values inclosed in parentheses are for information only.

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TABLE III. Test settings at 0° C

Pressure altitude (feet)	Indicated airspeed (knots)	True airspeed (knots)
0	150*	145
	200	193
	250*	241
	300	287
	350*	333
	400	379
	450*	423
	(500)	(466)
5,000	100	106
	150*	159
	200	211
	250*	262
	300	312
	350*	362
	400	410
	(450)	(456)
10,000	100	117
	150*	175
	200	231
	250*	287
	300	340
	350*	393
	400	443
15,000	100	129
	150*	192
	200	254
	250*	314
	300	372
	350*	427

*See paragraphs titles "Friction" and "Position error" (see 4.5.1.4 and 4.5.2.10).
Values inclosed in parentheses are for information only.

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TABLE IV. Test settings at -20° C

Pressure altitude (feet)	Indicated airspeed (knots)	True airspeed (knots)
0	150*	140
	200	186
	250*	232
	300	277
	350*	321
	400	364
	450*	407
	500	448
5,000	100	102
	150*	153
	200	203
	250*	253
	300	301
	350*	348
	400	394
	450*	439
10,000	100	113
	150*	168
	200	223
	250*	276
	300	328
	350*	378
	400	427
	(450)	(474)
15,000	100	124
	150*	185
	200	244
	250*	302
	300	358
	350*	411
	(400)	(462)
20,000	100	137
	150*	204
	200	269
	250*	331
	300	391
	350*	447

*See paragraphs titled "Friction" and "Position error" (see 4.5.1.4 and 4.5.2.10).
Values inclosed in parentheses are for information only.

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TABLE V. Test settings at -40° C

Pressure altitude (feet)	Indicated airspeed (knots)	True airspeed (knots)
0	150*	134
	200	179
	250*	222
	300	265
	350*	308
	400	350
	450*	390
	500	430
5,000	150*	147
	200	195
	250*	242
	300	289
	350*	334
	400	378
	450*	421
	(500)	(463)
10,000	100	108
	150*	161
	200	213
	250*	265
	300	315
	350*	363
	400	409
	(450)	(454)
15,000	100	119
	150*	177
	200	234
	250*	290
	300	343
	350*	394
	400	444
20,000	100	132
	150*	196
	200	258
	250*	318
	300	375
	350*	429
35,000	100	182
	150*	268
	200	349
	250*	423

*See paragraphs titled "Friction" and "Position error" (see 4.5.1.4 and 4.5.2.10).
Values inclosed in parentheses are for information only.

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TABLE VI. Test settings at -60° C

Pressure altitude (feet)	Indicated airspeed (knots)	True airspeed (knots)
0	150*	129
	200	171
	250*	213
	300	254
	350*	294
5,000	150*	141
	200	186
	250*	232
	300	276
	350*	319
10,000	100	103
	150*	154
	200	204
	250*	253
	300	301
15,000	100	114
	150*	170
	200	224
	250*	277
	300	328
20,000	100	126
	150*	187
	200	247
	250*	304
35,000	100	174
	150*	256

*See paragraphs titled "Friction" and "Position error" (see 4.5.1.4 and 4.5.2.10).

4.5.1.3.7 When required by the procuring activity, the indicator may be tested for scale error at any additional point or points on the scale, in combination with an indicated airspeed listed in the tables, and any pressure altitude and free air temperature. The errors at these additional test points shall not exceed the tolerances listed in the table.

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4.5.1.3.8 Scale error tolerances shall be:

(a) 75 percent of the test points shall be within ± 4 knots.
(b) 95 percent of the test points shall be within ± 6 knots.
(c) 100 percent of the test points shall be within ± 8 knots.

4.5.1.4 Friction. The indicator shall be tested for friction at every other asterisk-marked test point, beginning with the first test point indicated by an asterisk in tables I to VI, inclusive. The pressure shall be increased, to bring the pointer approximately to the desired reading, and then held constant while two readings are taken; the first before the indicator is tapped, and the second after the indicator is tapped. The difference between two such readings shall not exceed the tolerance specified in table VII. The pointer shall move smoothly while the pressure is varied uniformly, without vibration of the indicator. This test may be combined with the test for scale error.

4.5.1.5 Leakage.

4.5.1.5.1 Case leak. The pitot and static pressure connections of the indicator shall be jointed with a "Y" connection and shall be connected together, to the same mercury manometer and sources of suction and pressure. A suction of 15 inches Hg shall be applied to the pitot and static pressure connections of the indicator. With the source disconnected during a period of 10 seconds, the difference between the mercury levels in the manometer shall not change by more than 0.2 inch.

*4.5.1.5.2 Diaphragm capsule leak. The static connection shall be opened to the atmosphere and a pressure, sufficient to produce approximately full scale deflection of the pointer, shall be applied to the pitot connection of the indicator, at which point the connecting tubing shall be pinched off or otherwise completely sealed. During a period of one minute, the pointer shall not change its position by more than one knot.

4.5.2 Sampling tests. The sampling tests shall consist of the following tests (4.5.2.2 to 4.5.2.10) conducted on each sample specified in 4.5.2.1.

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TABLE VII. Tolerances

Test	Tolerances ± knots
Friction	5.0
Position error	5.0
Vibration error:	
Differential pressures at	
150 knots	4.0
350 knots	4.0
Vibration failure:	
Differential pressures at	
400 knots	5.0
Scale error:	
Seasoning	3.0
Overpressure	3.0
Capillary temperature 50°C	1.0
Capillary bending and strain	
bulb temperature 50°C	1.0

4.5.2.1 Sampling test instructions. Samples shall consist of three indicators selected at random by the inspector from each lot of 100 or less indicators which have passed the individual tests. A lot shall consist of identical indicators with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at substantially the same time. Indicators which have been subjected to the sampling tests shall not be delivered on contract until they have been rebuilt and resubmitted to all individual tests.

4.5.2.2 Weight. The indicator shall be weighed completely assembled and with connecting fittings, sunshields, and mounting screws. The weight of the indicator shall not exceed the value specified herein.

4.5.2.3 Magnetic property. The magnetic property of the indicator shall be determined in terms of the deflection of a free magnet, approximately 1-1/2 inches long, in a magnetic field with a horizontal intensity of 0.18, ± 0.01 gauss, when the indicator is held in various positions on an east-west line 5 inches from the center of the magnet. An aircraft compass, with compensating magnets removed therefrom, may be used as the free magnet for this test. The maximum deflection of the magnet shall not exceed 1 degree.

4.5.2.4 Low and high temperature operation.

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*4.5.2.4.1 Low temperature. The indicator shall be subjected to a temperature of -54° ± 2° C for 4 hours. At the end of this period and while still at the low temperature, the indicator shall meet the requirements specified for scale error in 4.5.1.3, except that only tables V and VI shall be used. Tolerances shall be as specified in 4.5.2.4.3.

*4.5.2.4.2 High temperature. The indicator shall be placed in a chamber and maintained at a temperature of 71° ± 2° C for 4 hours. While the indicator is still at this temperature, it shall meet the requirements specified for scale error as specified in 4.5.1.3, except that only tables I and II shall be used. Tolerances shall be as specified in 4.5.2.4.3.

4.5.2.4.3 Tolerances during low temperature and high temperature operation shall be:

(a) 75 percent of the test points shall be within ± 6 knots.
(b) 95 percent of the test points shall be within ± 11 knots.
(c) 100 percent of the test points shall be within ± 12 knots.

*4.5.2.5 Seasoning. The indicator shall be subjected to 1,000 applications of a differential pressure sufficient to produce approximately full-scale deflection. These applications shall be made at a rate not greater than 6 per minute. Not less than 1 hour following this test, the indicator shall be tested for scale errors as specified in 4.5.1.3 at 20° ± 2° C and 5,000 feet pressure altitude. The errors at the test points shall not exceed the tolerances specified in 4.5.1.3.8 by more than the tolerance specified in table VII.

4.5.2.6 Capillary temperature. A 3-foot length of capillary shall be subjected to a temperature as specified in table VII, with the indicator, bulb, and the remainder of the capillary at room temperature. The changes in reading, due to heating the 3-foot length of capillary, shall not exceed the tolerance specified in table VII.

4.5.2.7 Capillary bonding and strain. The indicator reading shall be obtained for a bulb temperature as specified in table VII. The capillary shall then be bent 20 times in one place on a radius of 1/2 inch through an angle of 90 degrees and straightened out after each bending. The flexible braided capillary covering, at points approximately 6 inches from the case and from the bulb, shall be so secured in a vise that the case and the bulb are suspended in the vise, to avoid crushing the covering. The change in reading as the result of bending and straining the capillary shall not exceed the tolerance specified in table VII.

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*4.5.2.8 Vibration error. Vibration error shall be in accordance with vibration test of paragraph 4.5.2.8.1. This test shall be conducted while the differential pressures specified in table VII are applied to the indicator. The indicator shall be subjected to vibration with an amplitude between 0.003 and 0.005 inch at frequencies from 500 to 3,000 cpm. The maximum amplitude of pointer vibration and the pointer variation from the original position shall not exceed the tolerances specified in table VII while the indicator is being vibrated.

*4.5.2.8.1 Vibration test procedure. The instrument shall be mounted on the appratus in its normal operating position. While being operated in accordance with paragraph 4.5.2.8, the item shall be vibrated with circular motion in a plane inclined 45 degrees to the horizontal plane with the following diameter of circular motion: 0.009 to 0.011 inch. The frequency of applied vibration shall be varied slowly from 5 to 50 cycles per second. No condition of applied vibration shall cause vibration errors involving pointer oscillation or variation in excess of those specified in table VII.

4.5.2.9 Overpressure.

4.5.2.9.1 A pressure of 15.0 inches Hg shall be applied to the pitot pressure connection of the indicator for a period of 10 minutes.

4.5.2.9.2 A pressure of 6.0 inches of water shall be applied to the static connection of the indicator for a period of 10 minutes.

*4.5.2.9.3 Not less than 5 minutes following the application of the overpressure tests, the indicator shall be tested for scale error as specified in 4.5.1.3 at +20°C and 5,000 feet pressure altitude. The errors at the test points shall not exceed the tolerances specified in table VII.

4.5.2.10 Position error. The indicator shall be observed while the indicator is held in any desired position and tapped. This reading shall not differ from the reading when the indicator is held in any position by more than the tolerances specified in table VII. This test shall be made at every other one of the points indicated by an asterisk in tables I to VI, inclusive.

*4.5.3 Special tests. Three samples representing the manufacturer's contract or order shall be subjected to the following tests.

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*4.5.3.1 Low temperature exposure. Low temperature exposure shall be in accordance with low temperature test of paragraph 4.5.3.1.1. While still at the specified temperature, pressure shall be applied to the pitot connector. The indicator shall operate without sticking. After the indicators has returned to room temperature, it shall meet the requirements specified in 4.5.1.3. There shall be no damage which would adversely affect subsequent operation.

*4.5.3.1.1 Low temperature test procedure. The equipment shall be placed within the test chamber and the chamber cooled to and maintained at a temperature -62°C (-80°F) for a period of 72 hours, at which time the equipment shall be inspected for conditions described in 4.5.3.1. The temperature of the chamber shall then be raised to -54°C (-65°F) and maintained for an additional 24-hour period or until stabilization of equipment temperature is reached, whichever is the longer. At the conclusion of this exposure period and while at this temperature, the equipment shall be inspected and operated in accordance with 4.5.3.1. The test item temperature shall then be returned to that of standard conditions and the item operated and inspected visually as specified in 4.5.3.1.

*4.5.3.2 High temperature exposure. The indicator shall be subjected to a temperature of 71° + 2° C for 24 hours. After the indicator has returned to room temperature, it shall meet the requirements specified in 4.5.1.3. There shall be no damage which would adversely affect subsequent operation.

*4.5.3.3 High altitude. The altitude test shall be in accordance with the procedure of paragraph 4.5.3.3.1. There shall be no damage as a result of this test.

*4.5.3.3.1 High altitude test procedure. The equipment shall be placed in the test chamber and the internal chamber temperature reduced to -54°C (-65° F). The internal pressure of the chamber shall be reduced to Condition A, B, C, D, E, and F below.

Condition	Pressure (In. Hg.)	Altitude (Ft.)
A	23.98	6,000
B	20.58	10,000
C	3.44	50,000
D	1.32	70,000
E	0.82	80,000
F	0.32	100,000

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The equipment shall be maintained under each conditions for 1 hour. At the conclusion of each period and while altitude and temperature conditions are maintained, the equipment shall be operated for 1 hour. No damage shall result from this test and the tolerances specified for individual tests shall apply.

*4.5.3.4 Acceleration. The indicator shall be mounted and accelerated first in its vertical axis and then in each of two axes that are perpendicular to the vertical axis and to each other. The indicator shall be subjected to an acceleration of 20g in each of the above axes for 30 seconds. At the end of this acceleration, the indicator shall meet the requirements specified in 4.5.1.3. No damage to the indicator shall result from this test.

*4.5.3.5 Vibration failure. Vibration failure shall be in accordance with vibration test of paragraph 4.5.3.5.1. The test shall be conducted while the differential pressure specified in table VII is applied to the indicator. The difference between the indication of the pointer at the beginning and at the end of the 3-hour vibration period shall not exceed the tolerance specified in table VII, and no looseness in the mechanism nor damage shall result in any part of the indicator.

*4.5.3.5.1 Vibration test procedure. This procedure constitutes a vibration failure test to detect faulty constructional details in panel-mounted aircraft instruments of reciprocating engine type aircraft. The instrument shall be mounted on the apparatus in its normal operating position. While being operated in accordance with this specification, the instrument shall be vibrated with circular motion of 0.018 to 0.020 inch diameter in a plane inclined 45 degrees to the horizontal plane and the frequency of vibration shall be varied uniformly from 5 to 50 cycles per second and return once each hour for a 3-hour period. At the completion of this test, the instrument shall be operated and a visual inspection shall be made to determine that the mechanism has not been damaged and operates normally.

*4.5.3.6 Salt spray. The indicator shall be subjected to a salt spray in accordance with salt spray test procedure of paragraph 4.5.3.6.1. At the end of the 50-hour period, the individual tests shall be repeated. No damage shall result from this test and the tolerances specified for the individual tests shall apply.

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*4.5.3.6.1 Salt spray test procedure. The procedure shall be in accordance with Appendix B of this specification. The length of the salt spray test shall be not less than 168 hours. At the end of the test period, the equipment shall be operated and inspected within 1 hour. Salt deposits resulting from the exposure conditions may be removed by rinsing with tap water prior to operation. Equipment shall again be operated and inspected for normal conditions and tolerances 48 hours later.

*4.5.3.7 Humidity. The humidity test shall be in accordance with humidity test procedures of paragraph 4.5.3.7.1. The indicator shall be mounted in its normal operating position in the chamber. The complete procedure shall be repeated 15 times. There shall be no evidence of corrosion or rusting of the internal or external parts of the indicator.

*4.5.3.7.1 Humidity test procedure. The equipment shall be placed in the test chamber and set up to simulate installed conditions. The test chamber shall be vented to the atmosphere to prevent the build-up of pressure. Prior to the starting of the test period, the chamber temperature shall be between 20° and 38° C (68° and 100° F) with uncontrolled humidity. During the first 2-hour period, the temperature shall be gradually raised to 71° C (160° F) unless otherwise specified in detail test requirements. The temperature selected shall be maintained during the next 6-hour period. The velocity of the air throughout the test area shall not exceed 150 feet per minute. During the following 16-hour period, the temperature in the chamber shall be gradually reduced to 20° to 38° C (68° to 100° F) which constitutes one cycle. The relative humidity throughout the cycle shall be 95 percent. Steam or distilled water having a pH value between 6.5 and 7.5 at 25° C (77° F) shall be used to obtain the desired humidity. The cycle shall be repeated a sufficient number of times to extend the total time of the test to 240 hours (10 cycles) or as specified by the applicable equipment specification. At the conclusion of the 240-hour period, the equipment shall be returned to standard conditions. The equipment shall then be operated and inspected within 1 hour for normal operating conditions and tolerances.

*4.5.3.8 Fungus. The fungus test shall be in accoedance with fungus test procedures of paragraph 4.5.3.8.1. After the test, the indicator shall meet the requirements specified for individual tests in paragraph 4.5.1.3. There shall be no deterioration which would adversely affect the subsequent operation.

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*4.5.3.8.1 Fungus test procedure. Four groups of fungi are listed below and one species of fungus from each group shall be used. In preparing the spore suspension, distilled water (having a pH value between 5.8 and 7.2 at a temperature between 22° C and 32° C (72° F and 89° F) shall be utilized. Approximately 10 ml. of the solution shall then be introduced directly into a tube culture of the fungus and the spores brought into suspension by vigorous shaking or by gentle rubbing of the spore layer with an innoculating loop without disturbing the agar surface. This process shall be repeated for each species of fungus. The separate spore suspensions from the four types of fungi shall be mixed together to provide a composite suspension. Actively growing cultures between 7 to 21 days old after initial innoculation shall be used for the preparation of the spore suspension. After preparation, the spore suspension will not be kept for more than a 24-hour period at temperatures from 22° C to 32° C (72° F to 89° F), or not more than 48 hours at temperatures from 2° C to 7° C (35° F to 45° F). The equipment, including applicable external connections, shall be placed in a mold chamber described in Appendix A of this document, (reference paragraph 6.5.1) maintaining an internal temperature of 30° ± 2° C (86° ± 3.6° F) and a relative humidity of 95 ± 5 percent, and sprayed with the suspension of mixed spores. The test period shall be 28 days. At the end of the test, the parts shall be visually examined for normal conditions. The equipment shall be operated and inspected for normal conditions and tolerances.

Organism		
Group I	Chaetomium globosum	6205
	Myrothecium verrucaria	9095
Group II	Memenoniella echinata	11973
	Aspergillus niger	6275
Group III	Aspergillus tamarii	10836
	Aspergillus terreus	10690
Group IV	Penicillium citrinum	9849
	Penicillium ochrochloron	9112

NOTE: When ordering cultures, request the cultures by name and serial number. The serial numbers of the cultures remain the same regardless of source. One known source is listed below. Other sources are acceptable.

Source: American Type Culture Collection
12301 Parklawn Dr., Rockville, MD 20852

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4.5.3.9 Mounting lugs. The indicator case with the mechanism and cover glass removed shall be mounted face downward on the movable head of a suitable testing machine, with the diameter of the case in a horizontal plane so that the mounting lugs receive no support. A suitable pin shall be inserted through the hole in the mounting lug and attached to the pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for 1 minute to each lug along the lug axis and toward the rear of the instrument. The lug shall withstand the applied load without fracture.

4.5.4 Inspection and test. Tests for methods of preservation and packaging will be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

4.5.5 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced will be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

*5. PREPARATION FOR DELIVERY. Unless otherwise specified in the individual contract or order, the following shall apply:

*5.1 Preservation and packaging.

*5.1.1 Level A. (Reference definition in MIL-STD-794.) Indicators shall be packaged in accordance with Method IIC of MIL-P-116, one each in a unit box conforming to PPP-B-636, Weather-Resistant Class. Each unit shall be cushioned with MIL-P-26514 Polyurethane. Cushioning thickness, density and configuration shall insure shock is dampened below the maximum permissable, nonoperating, shock limitations established for the indicator. Completed packs shall be afforded the rough handling tests presecribed in MIL-STD-794 to confirm packaging reliability.

*5.1.2 Level C. (Reference definition in MIL-STD-794.) Indicators shall be packaged identically to Level A requirements, except Method III (MIL-P-116) applies in lieu of Method IIC and Domestic Class PPP-B-636 Box shall be used as the unit container.

5.2 Packing.

*5.2.1 Level A. Indicators packages as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, Weather-Resistant Class. Insofar as practicable shipping containers shall be uniform shape and size and of minimum tare and cube consistent with the protection required.

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*5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in PPP-B-636 Domestic Class Boxes. Other requirements specified in 5.2.1 shall apply.

*5.2.3 Level C. Packing shall be applied which affords adequate protection against physical and mechanical damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

*5.3 Marking.

*5.3.1 Container markings. Unit packages and shipping containers shall be marked in accordance with MIL-STD-129. Nomenclature shall be as follows:

"INDICATOR, TRUE AIRSPEED, TYPE M-1A"

6. NOTES

6.1 Intended use. The type M-1A true airspeed indicator is intended for use on aircraft to indicate from 1,000 feet below sea level to 35,000 feet above sea level and under free air temperature conditions from +40° to -60° C.

*6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Data requirements (see 3.2).
(c) Levels of packaging and packing (see 5.1 and 5.2).

6.3 Definitions.

6.3.1 Amplitude. Whenever the word "amplitude" is specified, it shall mean the extent of motion measured from one extreme to the opposite extreme.

6.3.2 True airspeed knots in the tables of this specification are calculated from the following equation:

$$V_{true} = 87.0114 \sqrt{T_o \left(\left(1 + \frac{dP}{P} \right)^{.2857} - 1 \right)}$$

Where: V_{true} = true airspeed in knots.
T_o = corrected temperature in degrees K (free air temperature)
$T_o = (273.2 + T) - 1.07 (V_{true})^2 \times 10^{-4}$
(where T is test temperature in degrees C).
P = pressure corresponding to given altitude.
dP = pressure corresponding to given indicated airspeed (same units as P)

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*6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is OCAMA/MMEO, Tinker Air Force Base, Oklahoma 73145 and information pertaining to qualification of products may be obtained from that activity.

*6.5 Cross reference information. References to MIL-E-5272 contained in previous editions of MIL-I-6447 have been deleted in this edition because MIL-E-5272 has been made inactive for new design. Test procedures contained in MIL-E-5272 that are needed in MIL-I-6447 have been transposed to this document with a minimum of changes. It is the intent of these changes that indicators qualified under MIL-I-6447B are acceptable under MIL-I-6447C; new tests are not required.

*6.5.1 Interchangeability information for mold chamber. Appendix A to this document contains information formerly covered by Section 3 of MIL-C-9452 which has been rescinded. Small changes have been made to update essential data. (Reference paragraph 4.5.3.8.1)

*6.5.2 Interchangeability information for salt spray test. Appendix B to this document contains information formerly contained in FED-STD-151, Procedure 811, which has been rescinded. Small changes have been made to update essential data. (Reference paragraph 4.5.3.6.1)

*6.5.3 Availability of documents. Copies of rescinded specifications such as MIL-E-5272, MIL-C-9452 or FED-STD-151, Procedure 811, may be obtained from:

Director
Navy Publications Printing Services
NPPSO-4ND
Philadelphia, PA 19120

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*6.6 Marginal indicia. The margins of this specification are marked to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

*6.7 International standardization. Certain provision (MS28036) of this specification are the subject of international standardization agreement, STANAG 3434. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:
Navy - AS
Air Force - 71

Preparing Activity:
Air Force - 71

Project Number:
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APPENDIX A

10. FUNGUS RESISTANCE TESTING CHAMBER

10.1 Scope. This appendix outlines the design, fabrication, and testing to provide a facility for the fungus resistance testing of aeronautical and associated equipment formerly covered by MIL-C-9452 which has been cancelled.

10.2 Requirements.

10.2.1 General. The facility shall consist of an insulated chamber lined with a corrosion, moisture, and fungus-proof material such as stainless steel. The shelves shall be removable and fabricated from perforated stainless steel or wire capable of supporting 75 pounds per square foot. A humidifier, heat source, and other equipment necessary to meet the requirements outlined in this appendix and in Specification MIL-E-6447 regarding fungus resistance testing shall be provided. The facility shall be constructed so that it can be readily cleared and sterilized.

10.2.2 Specifications and standards. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

10.2.2.1 Standard parts. AN and MIL standard parts shall be used whenever they are suitable for the purpose and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

10.2.3 Design. This fungus resistance testing chamber shall be a self-contained unit capable of establishing, maintaining, and withstanding the environmental conditions specified under Fungus Resistance Tests of Specification MIL-I-6447. The facility shall be designed to meet the performance requirements outlined in 10.2.5.

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10.2.3.1 The chamber interior shall be completely fungus-proof and the chamber bottom shall be leak-proof and shall contain provisions for draining.

10.2.3.2 The doors shall seal tightly without a center mullion which would limit the capacity of the chamber.

10.2.3.3 The environmental conditions specified under Fungus Resistance Tests of Specification MIL-I-6447 should be maintained by using electrical heaters immersed in distilled water in the bottom of the chamber. The heaters must be of sufficient capacity to raise the chamber temperature from 65° F to 86° F in 10 minutes and must be automatically controlled.

10.2.3.4 A temperature indicator must be provided. Provisions for continuous recording of temperature and humidity will be provided.

10.2.3.5 The chamber shall operate on 110-volt, 60-cycle alternating current supplied through a single plug-in extension carried with the facility.

10.2.3.6 The velocity of air throughout the test chamber test area shall not be in excess of 25 feet per minute.

10.2.3.7 A minimum air velocity of 900 feet per minute is required to pass over the wet bulb to give accurate reading of the wet bulb temperature.

10.2.3.8 Figure 1 is for reference only and in no way limits or restricts the freedom of the contractor in designing to meet the requirements specified herein. The facility shall generally conform to Figure 1 and provide all of the features indicated therein. This does not exclude the use of a chest type (top opening) chamber.

10.2.4 Construction. The chamber shall be constructed in accordance with the highest grade shop and construction practices.

10.2.5 Performance. This chamber shall be capable of establishing, maintaining, and withstanding all environmental conditions specified under Fungus Resistance Tests of Specification MIL-I-6447, paragraph 4.5.3.8.1.

10.2.6 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL-STD-130.

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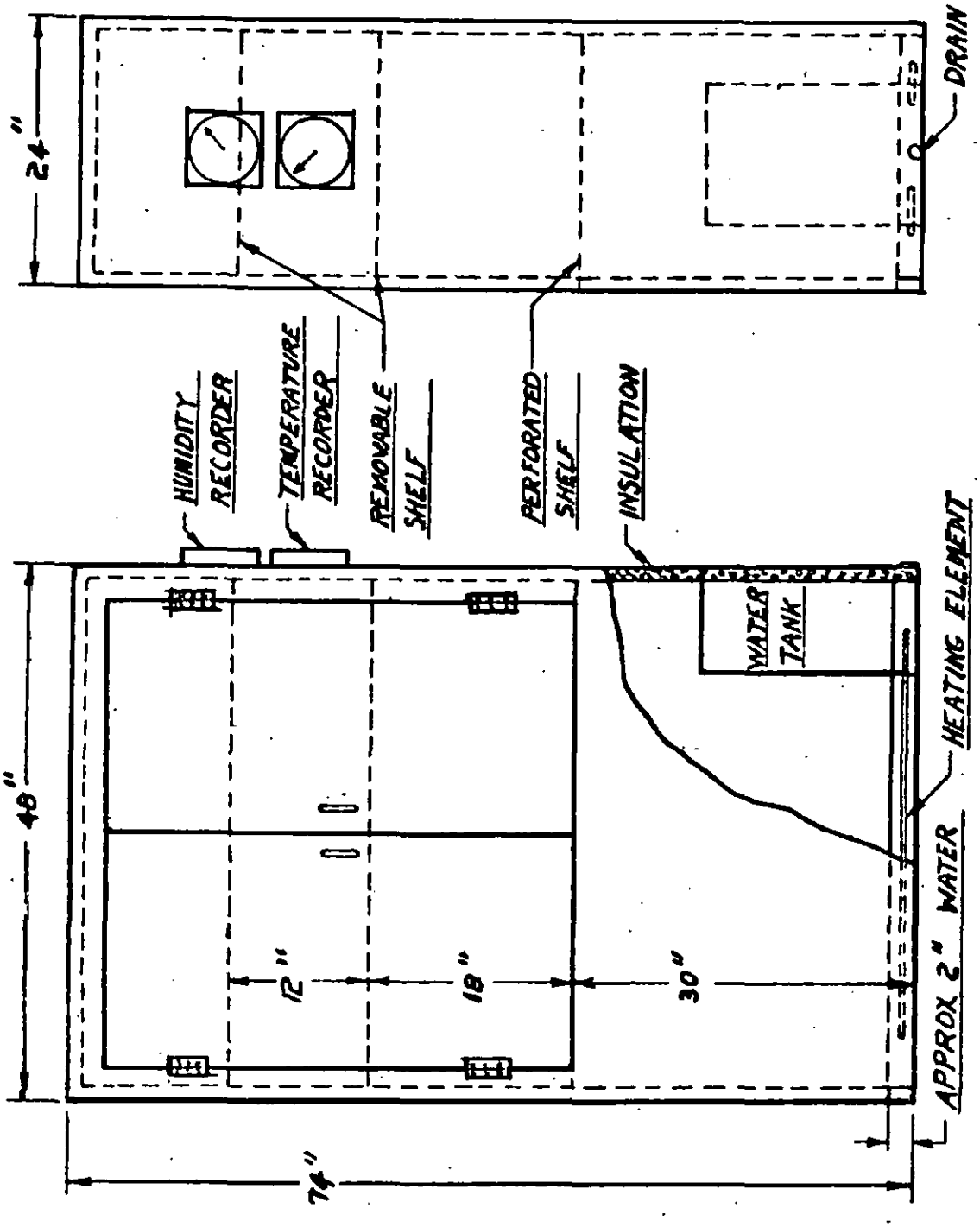


FIGURE 1

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

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